

Notation

This page is a reference, not a section: there is nothing to work through here, and it carries no exercises and no quick checks. Use it when you meet a symbol and want to know what it means and where the book fixed it. The symbols are grouped by the kind of thing they name, not by chapter, because a reader arrives with a symbol and not with a chapter number. Every entry links to the place where the book defines it; follow the link for the hypotheses, which a one-line gloss cannot carry.

Read the conventions first. Most of what looks like a clash between two symbols is resolved by one of them.

23.-1.1 The standing conventions

The field. F is an arbitrary field throughout, and $\mathbb{F}_p$ is the field with p elements (Definition 0.7.1). Chapters 10 to 12 and much of Part V need $\mathbb{R}$ or $\mathbb{C}$, and say so where they do. Vector spaces are V, U, W , plain italic, and are finite-dimensional unless the statement says otherwise.

Bold is for matrices and vectors, never for maps or spaces. A matrix is a bold upright capital, $\mathbf{A}$; a vector is a bold lowercase letter, $\mathbf{v}$; an operator or linear map is plain italic, T ; a space is plain italic, V . Entries stay plain: a_{ij} , or $(\mathbf{A})_{ij}$. A Greek letter naming a matrix is bold too, as $\mathbf{\Lambda}$ or $\mathbf{\Sigma}$.

Maps and composition. Linear maps are $T, S: V \rightarrow W$, never f . Composition is written ST , and it means: first T , then S . For functions in general the book writes $g \circ f$.

Coordinates. $[\mathbf{v}]_{\mathcal{B}}$ is the coordinate vector of $\mathbf{v}$ in the ordered basis $\mathcal{B}$ (Definition 1.5.2), and $[T]_{\mathcal{B}}^{\mathcal{C}}$ is the matrix of T with input basis $\mathcal{B}$ and output basis $\mathcal{C}$ (Definition 3.6.1). The subscript is the basis read into the first slot and the superscript the basis read into the second: for a map that is the input basis below and the output basis above, and for a bilinear form $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ it is the basis used on the first argument below and the one used on the second above. Both slots are always written out, even when they hold the same basis. A basis is an **ordered** list, written in parentheses.

Transpose and adjoint. $\mathbf{A}^{\top}$ is the transpose (Definition 0.9.10) and $\mathbf{A}^*$ the conjugate transpose (Definition 10.1.2), which is also the symbol for the adjoint of an operator (Definition 10.6.1). The prime T' is the dual map (Definition 4.4.1), and never a derivative of a map.

Inner products are linear in the first slot and conjugate-linear in the second (Definition 10.1.1). Over $\mathbb{R}$ the distinction disappears.

Ordering of eigenvalues and singular values. For Hermitian $\mathbf{A}$, the eigenvalues are listed **decreasing**, $\lambda_1(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A})$, with multiplicity. Singular values are listed decreasing as well, $\sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \dots \geq 0$ (Definition 12.8.1). Where a proof wants the increasing order it says so.

Indices start at 1, with three announced exceptions: Chapter 11 §09, where circulant and Fourier subscripts run from 0 to $n-1$ and are read modulo n ; Chapter 21 §§06 to 10, where the coordinates of F^{n+1} run from 0 so that $\mathbb{P}^n(F)$ has dimension n (Definition 21.6.2); and Chapter

22 §09, which follows Chapter 11 §09 for group elements and characters.

Defining, not asserting. $:=$ means “is defined to be”. A plain $=$ asserts an equality that has been or will be proved.

23.-1.2 Numbers, sets and fields

Symbol	Meaning	Defined at
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	natural numbers (with 0), integers, rationals, reals, complex numbers	Definition 0.3.1
$F, \mathbb{F}_p$	a field; the field with p elements	Definition 0.7.1
$\text{char } F$	characteristic of a field: the least $n > 0$ with $n \cdot 1 = 0$, or 0	Definition 0.7.2
$\bar{z}, z $	complex conjugate and modulus	Definition 0.6.2
$\{x \in S : P(x)\}$	set-builder notation, with a colon	Definition 0.3.2
$\emptyset, S \subseteq T$	the empty set; inclusion	Definition 0.3.3, Definition 0.3.4
$S \cup T, S \cap T, S \setminus T, S^c$	union, intersection, difference, complement	Definition 0.3.6, Definition 0.3.7, Definition 0.3.8
$S \times T, (x_1, \dots, x_n)$	Cartesian product and n -tuples	Definition 0.3.10, Definition 0.3.11
$\mathcal{P}(S)$	power set, the set of all subsets	Definition 0.3.12
$f: X \rightarrow Y,$ $x \mapsto f(x)$	a function and its rule	Definition 0.4.1
$f(A), f^{-1}(B)$	image of a subset, preimage of a subset	Definition 0.4.2
$g \circ f, \text{id}_X, f^{-1}$	composition, identity function, inverse function	Definition 0.4.4, Definition 0.4.5, Definition 0.4.7
$x \sim y, [x], X/\sim$	equivalence relation, equivalence class, quotient set	Definition 0.5.2, Definition 0.5.3, Definition 0.5.5
$S_n, \text{sgn}(\sigma), \text{inv}(\sigma)$	symmetric group; sign of a permutation; number of inversions	Example (Groups), Definition 6.3.2, Definition 6.3.1
A_n	alternating group, the even permutations	Definition 6.3.3
$F[x], F[x]_{\leq n}$	polynomials over F ; those of degree at most n	Definition 0.8.2, Definition 0.8.4
$\deg p, p(c)$	degree; evaluation at a scalar	Definition 0.8.3, Definition 0.8.6
$p \mid q, \gcd(p, q), \text{lcm}(p, q)$	divisibility, greatest common divisor, least common multiple in $F[x]$	Definition 5.1.1, Definition 5.2.2, Definition 5.2.4
$\langle p \rangle$	the ideal of $F[x]$ generated by p : all multiples of p	Definition 5.2.1
$\text{mult}_c(f)$	multiplicity of c as a root of f	Definition 5.4.1
p'	formal derivative of a polynomial	Definition 5.4.3
ℓ_i	Lagrange basis polynomial, $\ell_i(c_j) = \delta_{ij}$	Definition 5.6.1

23.-1.3 Spaces

Symbol	Meaning	Defined at
V, U, W	vector spaces over F ; U is usually a subspace	Definition 1.1.1, Definition 1.2.1
F^n, F^S $F^{(S)}$	column vectors; all functions $S \rightarrow F$ the free vector space on S : the finitely supported functions $S \rightarrow F$	Definition 1.1.1 Definition 14.2.3
$M_{m \times n}(F), M_n(F)$	matrices over F	Definition 0.9.1
$\text{span}(S)$	span; the span of the empty set is $\{\mathbf{0}\}$	Definition 1.3.2
$\dim V, \dim_F V$	dimension, over F when the field matters	Definition 1.6.1
$\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$	an ordered basis; $\mathcal{E}$ is the standard basis	Definition 1.5.1, Example (Standard Bases)
$[\mathbf{v}]_{\mathcal{B}}$	coordinate vector of $\mathbf{v}$ in $\mathcal{B}$	Definition 1.5.2
$U + W, U \oplus W$	sum of subspaces; internal direct sum	Definition 1.7.1, Definition 1.7.2
$V \times W$	product of vector spaces (the external direct sum)	Definition 3.9.1
$\mathbf{v} + U, V/U$	coset; quotient space	Definition 3.9.2, Definition 3.9.3
$\text{codim } U$	codimension, $\dim V/U$	Definition 4.5.1
V^*, φ	dual space; a linear functional, always a Greek letter	Definition 4.1.2, Definition 4.1.1
$\varphi_1, \dots, \varphi_n$	the dual basis of $\mathcal{B}$	Definition 4.1.3
U^0	annihilator of U inside V^*	Definition 4.2.1
$V^{**}, \text{ev}_{\mathbf{v}}$	double dual; evaluation at $\mathbf{v}$	Definition 4.3.1
$V_{\mathbb{C}}$	complexification of a real space	Definition 1.9.2
$\mathcal{F}$	a family of operators in $\mathcal{L}(V)$	Definition 8.9.1

23.-1.4 Vectors, matrices and their operations

Symbol	Meaning	Defined at
$\mathbf{v}, \mathbf{u}, \mathbf{w}, \mathbf{x}, \mathbf{y}$	vectors: bold lowercase	Definition 1.1.2
$\mathbf{0}, \mathbf{1}$	the zero vector; the all-ones vector	Definition 1.1.2, Definition 8.11.1
$\mathbf{e}_1, \dots, \mathbf{e}_n$	the standard basis of F^n	Example (Standard Bases)
$\mathbf{E}_{ij}$	matrix unit: 1 in entry (i, j) , zeros elsewhere	Example (Standard Bases)
$\mathbf{A}, \mathbf{B}, \mathbf{C}, a_{ij}$	matrices, bold upright capitals; their entries, plain	Definition 0.9.1
$\mathbf{0}_{m \times n}, \mathbf{I}, \mathbf{I}_n$	zero matrix, plain (bold $\mathbf{0}$ as a block); identity matrix	Definition 0.9.2, Definition 0.9.3
δ_{ij}	Kronecker delta: 1 when $i = j$, else 0	Definition 0.9.3
$\text{diag}(d_1, \dots, d_n)$	the diagonal matrix with these diagonal entries	Definition 0.9.4

Symbol	Meaning	Defined at
$\mathbf{A}^\top, \mathbf{A}^*$	transpose; conjugate transpose	Definition 0.9.10, Definition 10.1.2
$\text{tr } \mathbf{A}$	trace, the sum of the diagonal entries	Definition 0.9.12
$\det \mathbf{A}$	determinant	Definition 6.4.1
$\text{adj } \mathbf{A}$	adjugate, the transposed matrix of cofactors	Definition 6.6.2
M_{ij}, C_{ij}	minor and cofactor $(-1)^{i+j} M_{ij}$	Definition 6.6.1
$\mathbf{A}^{-1}$	inverse of a non-singular matrix	Definition 0.9.13
$p(\mathbf{A})$	a polynomial evaluated at a matrix	Definition 0.9.14
$\mathbf{P}_{ij}, \mathbf{D}_i(c),$ $\mathbf{I} + c\mathbf{E}_{ij}$	the three elementary matrices: swap, scale, add	Definition 2.4.1
$\mathbf{P}_\sigma$	permutation matrix	Definition 2.6.2
$\mathbf{A} = \mathbf{L}\mathbf{U},$ $\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}$	the LU and PLU factorizations	Definition 2.6.1, Definition 2.6.3
$\text{col}(\mathbf{A}), \text{row}(\mathbf{A}),$ $\text{nul}(\mathbf{A})$	column space, row space, null space	Definition 2.5.1, Definition 2.5.2, Definition 2.5.3
$\text{rank } \mathbf{A}, \text{nullity } \mathbf{A}$	rank and nullity of a matrix	Definition 2.5.4, Definition 2.5.5
$\mathbf{A} \sim \mathbf{B}$	similar: $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$	Definition 3.7.2
$\mathbf{A} \simeq \mathbf{B}$	congruent: $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$	Definition 13.2.1
$\mathbf{A} \approx \mathbf{B}$	equivalent over $F[x]$: $\mathbf{B} = \mathbf{U}\mathbf{A}\mathbf{V}$ with $\mathbf{U}, \mathbf{V}$ invertible over $F[x]$	Definition 9.7.2
equivalent matrices	$\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{P}$ with $\mathbf{P}, \mathbf{Q}$ non-singular	Definition 3.8.1
$\mathbf{A} \oplus \mathbf{B}$	block diagonal matrix with the given square diagonal blocks	Definition 7.1.2
$\mathbf{M}/\mathbf{A}, \mathbf{M}/\mathbf{D}$	Schur complement of an invertible block of a block matrix	Definition 7.3.1
$\mathbf{A} \otimes \mathbf{B}$	Kronecker product	Definition 7.4.1
$\text{vec } \mathbf{X}$	vectorization: the columns of $\mathbf{X}$ stacked into one column	Definition 7.4.2
$\mathbf{A} \circ \mathbf{B}$	Hadamard (entrywise) product	Definition 12.7.1
$[\mathbf{A}, \mathbf{B}] = \mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A}$	commutator	Definition 7.5.1
$\mathbf{J}$	the all-ones matrix, every entry 1	Example (Orthogonally diagonalizing a 3 by 3 matrix)
$\mathbf{A}'(t), \mathbf{x}'(t)$	entrywise derivative of a matrix- or vector-valued function of a real variable	Definition 15.6.1
$e^{\mathbf{A}}, f(\mathbf{A})$	matrix exponential; a function of a matrix through its Jordan form	Definition 9.9.2, Definition 9.9.1
$\lim_m \mathbf{A}^m$	entrywise convergence of matrices	Definition 9.10.1
$\mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{A}\mathbf{x} = \mathbf{0}$	a linear system and its homogeneous version	Definition 2.1.1, Definition 2.1.2
echelon form	the staircase shapes elimination produces	Definition 2.2.3, Definition 2.2.4

23.-1.5 Maps and their spaces

Symbol	Meaning	Defined at
$T, S: V \rightarrow W$	linear maps; ST means first T , then S	Definition 3.1.1
$T_{\mathbf{A}}$	the map $F^n \rightarrow F^m$, $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$	Example (Multiplication by a Matrix)
$\ker T$, $\operatorname{im} T$	kernel and image	Definition 3.2.1, Definition 3.2.2
$\operatorname{rank} T$, nullity T	$\dim \operatorname{im} T$ and $\dim \ker T$	Definition 3.3.2, Definition 3.3.1
$\mathcal{L}(V, W)$, $\mathcal{L}(V)$	all linear maps $V \rightarrow W$; the operators on V	Definition 3.4.1
$\operatorname{End}(V)$	the same space $\mathcal{L}(V)$, named as an algebra under composition	Definition 22.1.1
$p(T)$, id_V	a polynomial in an operator; the identity operator	Definition 3.4.2, Definition 3.1.1
$[T]_{\mathcal{B}}^{\mathcal{C}}$	matrix of T from the input basis $\mathcal{B}$ to the output basis $\mathcal{C}$	Definition 3.6.1
$[\operatorname{id}]_{\mathcal{B}}^{\mathcal{C}}$	change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$	Definition 3.7.1
$V \cong W$	isomorphic vector spaces	Definition 3.5.2, Definition 3.5.3
T'	dual map $W^* \rightarrow V^*$; the star is kept for adjoints	Definition 4.4.1
$\operatorname{tr} T$, $\det T$	trace and determinant of an operator	Definition 3.10.2, Definition 6.5.1
$T _U$, $\bar{T}$	restriction of T to an invariant subspace; the operator it induces on V/U	Definition 8.1.1, Definition 8.1.2
$\operatorname{GL}_n(F)$, $\operatorname{SL}_n(F)$	general and special linear groups	Example (Groups), Theorem 6.5.9
$\mathcal{M}(V_1, \dots, V_k; W)$	multilinear maps $V_1 \times \dots \times V_k \rightarrow W$; the semicolon separates inputs from output	Definition 14.1.2

23.-1.6 Inner products and norms

Symbol	Meaning	Defined at
$\langle \mathbf{u}, \mathbf{v} \rangle$	inner product: linear in the first slot , conjugate-linear in the second	Definition 10.1.1
$\langle \mathbf{A}, \mathbf{B} \rangle = \operatorname{tr}(\mathbf{B}^* \mathbf{A})$	the Frobenius inner product on matrices	Definition 10.1.1
$\ \mathbf{v}\ $	norm; the induced norm $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ unless said otherwise	Definition 10.1.3, Definition 15.1.1
$\mathbf{u} \perp \mathbf{v}$, θ	orthogonal vectors; the angle between two non-zero real vectors	Definition 10.1.4, Definition 10.1.5
$\mathbf{G}$	Gram matrix of a list, $(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$	Definition 10.2.3
$U^\perp$	orthogonal complement	Definition 10.3.1
P_U , $d(\mathbf{v}, U)$	orthogonal projection onto U ; distance to U	Definition 10.3.2, Definition 10.3.3
T^*	adjoint of an operator	Definition 10.6.1

Symbol	Meaning	Defined at
self-adjoint, normal	$T^* = T$; $T^*T = TT^*$	Definition 10.6.2, Definition 10.6.3
$O(n)$, $SO(n)$, $U(n)$	orthogonal, special orthogonal and unitary groups	Definition 10.7.3
$\mathbf{H}_w$, $\mathbf{G}(i, j; c, s)$	Householder reflection; Givens rotation	Definition 10.8.1, Definition 10.8.2
P_n , T_n	Legendre and Chebyshev polynomials	Definition 10.10.1, Definition 23.7.4
$\ \mathbf{x}\ _p$	the p -norm; q is the conjugate exponent, $1/p + 1/q = 1$	Definition 15.1.1, Corollary 17.11.5
$\ f\ _\infty$	sup norm on $C[0, 1]$	Definition 15.1.1
$B_{\ \cdot\ }$, B_1 , B_2 , B_∞	closed unit ball of a norm; the balls of the three p -norms	Definition 15.1.2
$\ \mathbf{A}\ _F$	Frobenius norm, $\sqrt{\text{tr}(\mathbf{A}^*\mathbf{A})}$; the subscript is obligatory	Lemma 11.11.1
$\ \mathbf{A}\ $, $\ \mathbf{A}\ _1$, $\ \mathbf{A}\ _\infty$	operator norm induced by a vector norm; largest column sum; largest row sum	Definition 15.3.2
$\kappa(\mathbf{A}) = \ \mathbf{A}\ \ \mathbf{A}^{-1}\ $	condition number	Definition 15.8.1
$\ \mathbf{y}\ _*$	dual norm, $\max_{\ \mathbf{x}\ \leq 1} \langle \mathbf{x}, \mathbf{y} \rangle$	Definition 17.4.4
$\ \mathbf{A}\ $, $\ \mathbf{A}\ _{(k)}$	unitarily invariant norm; the Ky Fan k -norm $\sigma_1 + \cdots + \sigma_k$	Definition 20.4.1, Definition 20.4.2
Φ , Φ^*	symmetric gauge function and its dual, with $\ \mathbf{A}\ = \Phi(\sigma(\mathbf{A}))$	Definition 20.3.1, Definition 20.3.3
$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}}$, $\ \mathbf{x}\ _{\mathbf{A}}$	the inner product $\mathbf{y}^* \mathbf{A} \mathbf{x}$ of a positive definite $\mathbf{A}$, and its energy norm	Theorem 12.4.2, Definition 23.7.1
$\Theta(U, W)$, $\theta_1 \leq \cdots \leq \theta_k$	principal angles between two subspaces, in $[0, \pi/2]$	Definition 19.8.1

23.-1.7 Bilinear forms, tensors and exterior algebra

Symbol	Meaning	Defined at
$\beta: V \times V \rightarrow F$, $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$	bilinear form; the quadratic form it carries	Definition 13.1.1, Definition 13.3.2
$[\beta]_{\mathcal{B}}^{\mathcal{B}}$	Gram matrix of a form, $\beta(\mathbf{v}_i, \mathbf{v}_j)$, indices uncrossed; $\mathcal{B}$ is written twice because each slot of the form is given a basis	Definition 13.1.2
$\text{rad}(\beta)$	radical of a form; β is non-degenerate exactly when it is $\{\mathbf{0}\}$	Definition 13.1.3, Definition 13.1.4
discriminant	the square class of $\det[\beta]_{\mathcal{B}}^{\mathcal{B}}$ in $F^\times$, the same for every basis	Definition 13.2.2
(n_+, n_-, n_0)	inertia of a real symmetric form; the signature is the pair (n_+, n_-)	Definition 13.5.1
$\text{In}(\mathbf{A})$	inertia triple of a Hermitian matrix, by sign of eigenvalue	Definition 16.10.1
$\mathbf{H}_f(\mathbf{a})$	Hessian matrix of a twice-differentiable function at a point	Definition 13.6.2

Symbol	Meaning	Defined at
isotropic, totally isotropic	$q(\mathbf{v}) = 0$ with $\mathbf{v} \neq \mathbf{0}$; a subspace on which β vanishes	Definition 13.7.1, Definition 13.7.2
hyperbolic plane, $\text{ind}(\beta)$	the plane of a hyperbolic pair; the Witt index	Definition 13.7.3, Definition 13.7.4
$U \perp W$	orthogonal direct sum for a form: the summands pair to zero	Definition 13.7.3
$(V, \beta) \cong (V', \beta')$	isometric bilinear spaces (this use of $\cong$ is local to forms)	Definition 13.8.1
$\text{Isom}(\beta)$	isometry group of a form, $\{\mathbf{P} : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}\}$	Definition 13.10.1
$O(p, q)$, $\mathbf{I}_{p,q}$, $\text{Sp}(2m, F)$, $\mathbf{\Omega}_{2m}$	the indefinite orthogonal and symplectic groups and their standard matrices	Definition 13.10.2
$V \otimes W$, $\mathbf{v} \otimes \mathbf{w}$	tensor product and a simple tensor	Definition 14.2.1, Definition 14.2.2
$S \otimes T$, $\mathcal{B} \otimes \mathcal{C}$	tensor product of maps; the product basis, first factor slow	Definition 14.3.1, Definition 14.3.2
Θ	the isomorphism $V^* \otimes W \rightarrow \mathcal{L}(V, W)$	Theorem 14.3.9
C , C_b^a	contraction $V^* \otimes V \rightarrow F$; contraction of one upper against one lower slot	Definition 14.4.1, Definition 14.4.3
$T_q^p(V)$, $T_{j_1 \dots j_q}^{i_1 \dots i_p}$	mixed tensors of type (p, q) and their components	Definition 14.4.2, Definition 14.5.1
$V^{\otimes k}$, $\text{T}(V)$	k -th tensor power, with $V^{\otimes 0} = F$; the tensor algebra	Definition 14.6.1, Definition 14.6.2
$\text{Sym}^k V$, $\text{Sym}(V)$	k -th symmetric power; the symmetric algebra, written by juxtaposition	Definition 14.7.2, Definition 14.7.3
$(V^{\otimes k})^{S_k}$	the symmetric tensors, fixed by every permutation of the slots	Definition 14.7.5
$F[x_1, \dots, x_n]$	the polynomial algebra in n variables	Definition 14.7.4
$\Lambda^k V$, $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$	k -th exterior power and a wedge	Definition 14.8.2
ΛV , $\mathbf{e}_S$	exterior algebra; its basis element for an increasing index set	Definition 14.10.1
$\Lambda^k T$	exterior power of a linear map; on the top power it is $\det T$	Definition 14.9.1
$\varepsilon_{\mathbf{v}}$	wedging by a fixed vector, $\omega \mapsto \mathbf{v} \wedge \omega$	Lemma 14.9.1
$\mathcal{I}$, $\mathcal{I}_q$	a two-sided ideal of an algebra; the one whose quotient is the Clifford algebra	Definition 14.10.3, Definition 13.12.2
$\mathbb{H}$	the quaternions	Definition 13.12.3

23.-1.8 Spectra and canonical forms

Symbol	Meaning	Defined at
$p_T(x) = \det(x\mathbf{I} - T)$	characteristic polynomial, monic	Definition 6.10.1, Definition 6.10.2
$m_T(x)$	minimal polynomial: the monic generator of the annihilating ideal	Definition 8.5.1, Definition 5.7.1
$\text{spec}(T)$	spectrum, the set of eigenvalues	Definition 8.2.3
$E_\lambda(T) = \ker(T - \lambda I)$	eigenspace	Definition 8.2.1, Definition 8.2.2

Symbol	Meaning	Defined at
$a_T(\lambda), g_T(\lambda)$	algebraic and geometric multiplicity of λ	Definition 8.3.1, Definition 8.3.2
$G_\lambda(T)$	generalized eigenspace, $\ker(T - \lambda I)^n$	Definition 9.1.1, Definition 9.1.2
$\mathbf{J}_k(\lambda)$	$k \times k$ Jordan block: λ on the diagonal, 1 on the super diagonal	Definition 9.3.1, Definition 9.3.2
$\mathbf{\Lambda}(\lambda), \mathbf{C}_k(\lambda)$	rotation-scaling block of a conjugate pair; the real Jordan block built from it	Definition 9.4.1
$\mathbf{C}(p)$	companion matrix of a monic p	Definition 9.5.3
$Z(\mathbf{v}; T), m_{T, \mathbf{v}}$	cyclic subspace generated by $\mathbf{v}$; the T -annihilator of $\mathbf{v}$	Definition 9.5.1, Definition 9.5.2
$d_1 \mid d_2 \mid \cdots \mid d_r$	invariant factors, with $d_r = m_T$; their prime powers are the elementary divisors	Definition 9.6.1
$\rho(\mathbf{A})$	spectral radius, the largest modulus in the spectrum	Definition 15.4.1
$\mathbf{\Pi}$	the limit $\lim_m \mathbf{A}^m$ of a convergent matrix	Theorem 9.10.4
$\mathbf{A} \succeq 0, \mathbf{A} \succ 0$	positive semidefinite, positive definite; Hermitian is implied	Definition 12.1.1
$\mathbf{A} \succeq \mathbf{B}$	the Loewner order, $\mathbf{A} - \mathbf{B} \succeq 0$	Definition 12.5.1
$\mathbf{A}^{1/2}$	the unique positive semidefinite square root	Theorem 12.2.1
$ \mathbf{A} = (\mathbf{A}^* \mathbf{A})^{1/2}$	absolute value of a matrix	Definition 12.2.1
$\sigma_1(\mathbf{A}) \geq \cdots \geq$	singular values, decreasing	Definition 12.8.1
$\sigma_{\min(m,n)}(\mathbf{A})$		
$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$	a singular value decomposition: $\mathbf{U}$ and $\mathbf{V}$ unitary, $\mathbf{\Sigma}$ the same size as $\mathbf{A}$ with the singular values on its diagonal and zeros elsewhere	Theorem 12.8.1
$\mathbf{A}^+$	Moore–Penrose pseudoinverse	Definition 12.11.1
$\lambda_1(\mathbf{A}) \geq \cdots \geq$	eigenvalues of a Hermitian matrix, decreasing;	Definition 16.1.1,
$\lambda_n(\mathbf{A}), \boldsymbol{\lambda}(\mathbf{A})$	the vector of them	Definition 16.8.1
$R_{\mathbf{A}}(\mathbf{x}), R_T(\mathbf{v})$	Rayleigh quotient $\langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$	Definition 16.1.1
$s_k(\mathbf{A})$	Ky Fan partial sum $\lambda_1(\mathbf{A}) + \cdots + \lambda_k(\mathbf{A})$	Corollary 16.6.5
$\mathbf{d}(\mathbf{A})$	the diagonal vector $(a_{11}, \dots, a_{nn})$ of a Hermitian matrix	Definition 16.8.1
$\mathcal{W} = (W_1, \dots, W_k)$	a flag of subspaces, and a list adapted to it	Definition 16.7.1
$\mathcal{H}(\mathbf{A})$	Hermitian dilation, whose eigenvalues are $\pm \sigma_i(\mathbf{A})$ and zeros	Proposition 16.9.3
$r_i(\mathbf{A}), D_i(\mathbf{A})$	Gershgorin radius and disc	Definition 19.1.1
$O_{ij}(\mathbf{A})$	Brauer oval of Cassini	Theorem 19.2.1
$W(\mathbf{A})$	numerical range, $\{\mathbf{x}^* \mathbf{A} \mathbf{x} : \ \mathbf{x}\ _2 = 1\}$	Definition 19.2.1
$\mathbf{E}, \tilde{\mathbf{A}}$	a perturbation, in $\mathbf{A} + \mathbf{E}$; a perturbed matrix when it is not of that shape	Definition 15.8.1
$\text{sep}_F(\mathbf{A}, \mathbf{B})$	separation of two matrices in the Frobenius norm	Definition 19.7.1
$\kappa_{\mathbf{A}}(\lambda)$	condition number of a simple eigenvalue	Definition 19.11.1
$\mathcal{S}_{\mathbf{A}, \mathbf{B}}$	Sylvester operator $\mathbf{X} \mapsto \mathbf{A} \mathbf{X} - \mathbf{X} \mathbf{B}$	Definition 11.10.1
$\mathbf{B} - \lambda \mathbf{A}$	a matrix pencil; its generalized eigenvalues solve $\mathbf{B} \mathbf{x} = \lambda \mathbf{A} \mathbf{x}$	Definition 12.4.1

23.-1.9 Orders, convexity and non-negative matrices

Symbol	Meaning	Defined at
$\mathbf{x} \geq \mathbf{0}$, $\mathbf{A} \geq 0$, $\mathbf{A} > 0$	the entrywise order; a positive matrix has every entry positive	Definition 18.1.1
$\mathbb{R}_{\geq 0}^m$	the non-negative orthant	Definition 17.5.1
$\mathbf{x} \prec \mathbf{y}$, $\mathbf{x} \prec_w \mathbf{y}$, $\mathbf{x}^\downarrow$	majorization, weak majorization, decreasing rearrangement	Definition 16.8.1
$\text{conv } S$, $\text{aff } S$	convex hull; affine hull, a coset of a subspace	Definition 17.1.4, Definition 17.1.5
$[\mathbf{x}, \mathbf{y}]$	the segment from $\mathbf{x}$ to $\mathbf{y}$	Definition 17.1.1
Δ_n	standard simplex: n non-negative weights adding to 1	Definition 17.1.2
$\text{int } (S)$, $\bar{S}$	interior and closure in a finite-dimensional real space	Definition 17.3.2, Definition 17.1.6
$P_C(\mathbf{x})$, $d(\mathbf{x}, C)$	nearest point of a closed convex set, and the distance to it	Theorem 17.3.1
p_K , h_K , K°	gauge, support function and polar set of a convex set	Definition 17.4.1, Definition 17.4.2, Definition 17.4.3
K^*	dual cone, inside V itself	Definition 17.5.3
$\text{cone}(\mathbf{a}_1, \dots, \mathbf{a}_m)$	finitely generated cone	Definition 17.5.2
$\text{epi } f$	epigraph, $\{(\mathbf{x}, s) : s \geq f(\mathbf{x})\}$	Definition 17.10.1
$\text{ext } C$	extreme points of a convex set	Definition 17.8.1, Definition 17.8.2
$\mathcal{D}_n$, Ω_n	real density matrices; doubly stochastic matrices	Definition 17.8.3, Definition 17.8.4
$v(\mathbf{A})$	value of the matrix game with payoff matrix $\mathbf{A}$	Definition 17.7.2, Definition 17.7.1
$\mathbf{A} \# \mathbf{B}$	geometric mean of two positive definite matrices	Definition 20.11.1
$L_f(t_1, \dots, t_n)$	Loewner matrix of divided differences	Definition 20.9.2, Definition 20.9.1
$G(\mathbf{A})$	directed graph of a square matrix: an edge $j \rightarrow i$ when $a_{ij} \neq 0$	Definition 18.2.1
irreducible, primitive	no proper invariant coordinate subspace; some power is positive	Definition 18.2.3, Definition 18.5.1
h , $\omega = e^{2\pi i/h}$	period of an irreducible non-negative matrix, and the root of unity it names	Definition 18.5.2
$\underline{r}_{\mathbf{A}}(\mathbf{x})$, $\bar{r}_{\mathbf{A}}(\mathbf{x})$	lower and upper Collatz–Wielandt functions; the bars are not conjugation	Definition 18.4.1
Perron root and vectors	$\rho(\mathbf{A})$, the right vector $\mathbf{v} > \mathbf{0}$ with $\mathbf{1}^\top \mathbf{v} = 1$, the left vector $\mathbf{w}$ with $\mathbf{w}^\top \mathbf{v} = 1$	Definition 18.1.2, Definition 18.3.1
$\mathbf{Q}$, $\mathbf{R}$, $\mathbf{N}$	transient block, absorbing block and fundamental matrix of an absorbing chain	Definition 18.6.1
$\mathbf{G}$, $\mathbf{H}$, α	Google matrix, raw link matrix, damping factor	Definition 18.8.2, Definition 18.8.1
$\mathbf{K}_n$	the second-difference matrix, a non-singular M-matrix	Definition 18.9.2, Definition 18.9.1

Symbol	Meaning	Defined at
stochastic matrix, steady state	column sums 1 and non-negative entries; a fixed probability vector	Definition 8.11.1, Definition 8.11.2

23.-1.10 Algebras, groups and representations

Symbol	Meaning	Defined at
$A, 1_A$	an associative algebra over F and its unit; plain italic, never bold	Definition 13.12.1
$\mathcal{T}_n, \mathcal{N}$	the upper triangular matrices in $M_n(F)$; its ideal of strictly upper triangular ones	Example (Six subalgebras)
$F\langle S \rangle$	the subalgebra generated by a set: the span of the words in S	Definition 22.1.4
$F[T], F[\mathbf{A}]$	the algebra of polynomials in one operator or matrix, of dimension $\deg m_T$	Proposition 22.1.2
$F[G], \delta_g$	group algebra of a finite group and its basis of point masses	Definition 22.1.5
L_a	left multiplication by a ; the regular representation is $a \mapsto L_a$	Theorem 22.1.3
$S', S'', Z(A)$	commutant, double commutant, center	Definition 22.1.6, Proposition 22.1.5
C_j, L_W, R_U	the column ideals of $M_n(F)$, and the left and right ideals of a subspace	Definition 22.2.1, Proposition 22.2.4
$A/\mathcal{I}, \pi$	quotient algebra by a two-sided ideal, and the quotient homomorphism	Definition 22.2.2
$\text{rad } A$	radical of an algebra: killed by every irreducible representation	Definition 22.3.5
$a\mathbf{v}$, an A -space	an action of A on V , that is a homomorphism $A \rightarrow \text{End}(V)$	Definition 22.3.2
simple, semisimple	an A -space with no proper non-zero invariant subspace; a direct sum of such	Definition 22.3.3, Definition 22.3.4
$\text{End}_A(V)$, $\text{Hom}_A(V, W)$	equivariant maps, $f(a\mathbf{v}) = af(\mathbf{v})$; the subscript is never dropped	Definition 22.4.1
$A(\mathcal{F})$	the subalgebra generated by a family of operators	Lemma 22.4.2
$\mathbf{v}\varphi$	the rank-one operator $\mathbf{x} \mapsto \varphi(\mathbf{x})\mathbf{v}$	Theorem 22.5.2
e_i	orthogonal idempotents of a product of algebras: italic, never bold	Definition 22.6.1
$\rho, \chi_\rho(g) = \text{tr } \rho(g)$	a representation of a group and its character	Definition 22.7.1, Definition 22.8.2
irreducible, equivalent	no proper non-zero subrepresentation; conjugate by one isomorphism	Definition 22.7.4, Definition 22.7.3
$\text{Cl}(G), \langle f, f' \rangle$	class functions on G , and their averaged inner product	Definition 22.8.3, Definition 22.8.4
V^G	the fixed space of a representation	Lemma 22.8.4
$\langle g \rangle, G \times H$	the cyclic subgroup generated by g ; a direct product of groups	Definition 22.9.1, Definition 22.9.2

Symbol	Meaning	Defined at
$\widehat{G}, \widehat{f}, \text{ev}_g$	dual group of a finite abelian group, Fourier transform, evaluation at g	Definition 22.9.3, Definition 22.9.5
$\mathbf{U}$	character matrix, $(\mathbf{U})_{mj} = \chi_j(g_m)/\sqrt{n}$, indices from 0	Definition 22.9.4
$\mathbf{C}, \mathbf{F}$	circulant matrix; the Fourier matrix	Definition 11.9.1, Theorem 11.9.3
$\mathbf{c} * \mathbf{d}$	cyclic convolution, subscripts read modulo n	Definition 11.9.1

23.-1.11 Geometry

Symbol	Meaning	Defined at
$\mathcal{A}, \mathcal{B}, \mathcal{C}$	affine spaces, plain calligraphic	Definition 21.1.1
$P, Q, R, \overrightarrow{PQ}, P + \mathbf{v}$	points of an affine space, not bold ; the vector between two points; translation of a point	Definition 21.1.1
$\mathbb{A}^n(F)$	affine n -space: F^n with its origin forgotten	Definition 21.1.1
θ_O	the choice-of-origin bijection $\mathcal{A} \rightarrow V, P \mapsto \overrightarrow{OP}$	Proposition 21.1.3
$\vec{f}, \widehat{f}$	linear part of an affine map; its block matrix, homogenizing coordinate last	Theorem 21.1.5, Proposition 21.1.7
$\text{Aff}(\mathcal{A}), \tau_{\mathbf{v}}$	the affine group, and translation by $\mathbf{v}$	Theorem 21.1.6
$\vec{A}, A \vee B, A \parallel B$	direction space of a flat; the join; parallelism	Definition 21.2.1, Definition 21.2.2, Definition 21.2.3
P_J	the face of a polyhedron cut out by making the rows $i \in J$ tight	Proposition 21.3.1, Definition 21.3.1
$E(n), t_{\mathbf{b}}, \text{Fix}(f)$	the Euclidean group, translation by $\mathbf{b}$, fixed set of a motion	Definition 21.4.1
$\mathbf{R}_{\theta}, \mathbf{M}_{\theta}$	plane rotation and plane reflection matrices	Theorem 10.7.5
$\widetilde{\mathbf{A}}, \tilde{\mathbf{x}}$	big matrix and big vector of a quadric, with $Q(\mathbf{x}) = \tilde{\mathbf{x}}^{\top} \widetilde{\mathbf{A}} \tilde{\mathbf{x}}$	Definition 21.5.1
$\mathbb{P}(V), \mathbb{P}^n(F)$	projective space of V , of dimension $\dim V - 1$; $\mathbb{P}(F^{n+1})$	Definition 21.6.1
$[\mathbf{x}] = [x_0 : \cdots : x_n]$	homogeneous coordinates; coordinates run from 0	Definition 21.6.2
$U_0, U_j, H_{\infty}, \iota$	the standard affine chart, the other charts, the hyperplane at infinity, the chart bijection	Theorem 21.6.2
$S \vee T, S \cap T$	join and meet of projective subspaces	Proposition 21.6.5, Definition 21.6.3
$[T], [\mathbf{A}]$	the projectivity induced by an isomorphism or an invertible matrix	Definition 21.7.1
$\text{PGL}(V), F^{\times}$	projective linear group; the multiplicative group of the field	Theorem 21.7.3
$(A, B; C, D)$	cross-ratio of four points: itself a point of $\mathbb{P}^1(F)$, not a scalar	Definition 21.7.3
S^0	projective annihilator, Chapter 4's U^0 read in $\mathbb{P}(V^*)$	Theorem 21.8.3, Definition 21.8.1

Symbol	Meaning	Defined at
$\mathbf{o}$	a chosen representative vector of a projective point	Lemma 21.9.1
$Z(q), P^\beta$	zero set of a quadratic form; the polar of a point with respect to a conic	Definition 21.10.1, Definition 21.10.2
$p_{ij}, [L], \mathcal{K}$	Plücker coordinates of a line, the point they form, and the Klein quadric	Definition 21.10.3
vol	volume: of a parallelepiped it is $ \det $; extended to k -dimensional volume	Definition 6.9.1, Definition 12.3.1
$(\text{In } \mathbf{A}, \text{In } \tilde{\mathbf{A}})$	the pair of inertia triples of a real quadric, an invariant of its affine equivalence class	Definition 21.5.2, Definition 16.10.1

23.-1.12 Computation

Symbol	Meaning	Defined at
$u, \text{fl}, \mathcal{R}, \delta$	unit roundoff, the rounding map, the representable numbers, one rounding	Definition 23.1.1
$\gamma_k = ku/(1 - ku)$	the accumulated-rounding constant	Lemma 23.1.1
$\hat{\mathbf{x}}, \Delta \mathbf{A}$	a computed object; a data perturbation produced by an algorithm	Definition 23.1.2, Definition 23.1.3
flop, g_n	one arithmetic operation; the growth factor of elimination	Definition 23.2.1, Definition 23.13.1
$\mathcal{K}_k(\mathbf{A}, \mathbf{b}), \theta_1, \dots, \theta_k$	Krylov space $\text{span}(\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{k-1}\mathbf{b})$; the Ritz values from it	Definition 23.6.1, Definition 23.6.2
$\mathbf{L}, \mathbf{L}_0, \mu_2(\mathbf{L})$	graph Laplacian $\mathbf{D} - \mathbf{A}$, the reduced Laplacian, the Fiedler value	Definition 23.11.1, Definition 23.11.2
$\mathbf{X}, \mathbf{X}_c, \mathbf{S}$	data matrix, its centered form, the sample covariance	Definition 23.10.1, Definition 23.10.2
$\mathbf{M}, \mathbf{K}, \omega_j$	mass matrix, stiffness matrix, natural frequencies	Definition 23.9.2
$\mathbf{W}_n, \mathbf{P}_n, \mathbf{D}_m, \mathbf{B}_n$	unnormalized transform matrix, even-odd permutation, twiddle matrix, butterfly	Definition 23.12.1, Definition 23.12.2
ω	the exponent of matrix multiplication	Definition 23.13.2

23.-1.13 Where the book overrides itself

Twenty-four chapters do not have twenty-four alphabets. Several letters carry more than one meaning, and a few English words carry more than one definition. In every case the local meaning is declared where it is used, and in every case there is a tell — an argument, a subscript, a typeface — that decides which one is meant. The rows below list the clashes worth knowing before you meet them.

Symbol or word	The meanings	How to tell
U	a generic upper triangular matrix (Definition 0.9.5, Definition 2.6.1); a unitary matrix, in the Schur and singular value factorizations (Corollary 11.1.3, Definition 12.8.1); the character matrix of a finite abelian group (Definition 22.9.4)	the chapter: triangular in Chapters 2 and 22, unitary from Chapter 11 on, the character matrix only in Chapter 22 §09
V	the right factor of a singular value decomposition, $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ (Theorem 12.8.1); the Vandermonde matrix $\mathbf{V}(x_1, \dots, x_n)$ (Theorem 6.8.1); one of the two matrices invertible over $F[x]$ in $\mathbf{B} = \mathbf{UAV}$ (Definition 9.7.2). Plain V is a vector space	the Vandermonde matrix always carries its nodes; the typeface for the space, and Chapter 9 §07 is the one place V is invertible only over $F[x]$
irreducible	a polynomial with no non-trivial factorization (Definition 5.3.1); a non-negative matrix with no invariant coordinate subspace (Definition 18.2.3); an A -space or a representation with no proper non-zero invariant subspace (Definition 22.3.3, Definition 22.7.4)	what the word is applied to. The three notions are unrelated
e_i versus e_i	italic e_i is an idempotent of a product of algebras (Definition 22.6.1); bold e_i is the i -th standard basis vector (Example (Standard Bases))	the typeface, and that is the whole reason Chapter 22 chose italic
G	Gram matrix (Definition 10.2.3); Givens rotation $\mathbf{G}(i, j; c, s)$ (Definition 10.8.2); Google matrix (Definition 18.8.2); the alternating Gram matrix of the Plücker form (Definition 21.10.3). Plain $G(\mathbf{A})$ is the directed graph of a matrix (Definition 18.2.1)	the Givens rotation always carries its four arguments; the graph is plain italic with its matrix in brackets
H, K	Hessian $\mathbf{H}_f(\mathbf{a})$ (Definition 13.6.2); Householder reflection $\mathbf{H}_w$ (Definition 10.8.1); the Hermitian and skew parts of a matrix (Lemma 19.2.3); the raw link matrix (Definition 18.8.1); the second-difference matrix $\mathbf{K}_n$ (Definition 18.9.2); the stiffness matrix (Definition 23.9.2). Calligraphic $\mathcal{K}_k(\mathbf{A}, \mathbf{b})$ is a Krylov space (Definition 23.6.1) and $\mathcal{K}$ the Klein quadric (Definition 21.10.3); blackboard $\mathbb{H}$ is the quaternions (Definition 13.12.3)	the Hessian and the Householder reflection always carry their argument, the second-difference matrix always its size K_n , the Krylov space its subscript and arguments
L	the lower triangular factor of an elimination (Definition 2.6.1); the graph Laplacian (Definition 23.11.1); the Lax matrix (Definition 23.8.1). Plain L_a is left multiplication in an algebra (Theorem 22.1.3) and L_f a Loewner matrix (Definition 20.9.2); calligraphic $\mathcal{L}(V, W)$ is a space of maps (Definition 3.4.1)	Chapter 23 §§08 and 11 each declare their use at the head of the section; the plain letters carry a subscript

Symbol or word	The meanings	How to tell
$\mathbf{R}, r$	the $\mathbf{R}$ of a QR factorization; the plane rotation $\mathbf{R}_\theta$ (Theorem 10.7.5); the rotation through $2\pi/n$ in Chapter 22; the absorbing block of a Markov chain (Definition 18.6.1). Plain $R_{\mathbf{A}}(\mathbf{x})$ is the Rayleigh quotient (Definition 16.1.1) and $r_i(\mathbf{A})$ a Gershgorin radius (Definition 19.1.1)	the Rayleigh quotient is plain italic and takes a vector; the Gershgorin radius is plain and takes an index
$\mathbf{Q}$	the orthogonal factor of a QR factorization; the orthogonal linear part of a Euclidean motion (Definition 21.4.1); the quarter turn in Chapter 22; the transient block of an absorbing chain (Definition 18.6.1). Blackboard $\mathbb{Q}$ is the rationals (Definition 0.3.1)	the chapter, and the blackboard face for the field
$\mathbf{S}, \mathcal{S}$	a congruence or a similarity matrix; a Schur complement in Chapter 23 §02; the sample covariance (Definition 23.10.2); the cyclic shift in Chapter 11 §09. Calligraphic $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$ is the Sylvester operator (Definition 11.10.1)	each section of Chapter 23 says which it means; the Sylvester operator carries both its matrices
$\mathbf{N}, \mathcal{N}$	the fundamental matrix $(\mathbf{I} - \mathbf{Q})^{-1}$ of an absorbing chain (Definition 18.6.1); an incidence matrix, with $\mathbf{L} = \mathbf{N}\mathbf{N}^\top$ (Definition 2.7.1); the strictly upper triangular matrices $\mathcal{N}$ (Example (Six subalgebras)). Blackboard $\mathbb{N}$ is the natural numbers	the calligraphic face for the algebra, the blackboard face for the number set
$\mathbf{J}$	the all-ones matrix (Example (Orthogonally diagonalizing a 3 by 3 matrix)); the Jordan block $\mathbf{J}_k(\lambda)$ (Definition 9.3.1)	the Jordan block always carries its size and its eigenvalue
$\mathbf{C}$	the companion matrix $\mathbf{C}(p)$ (Definition 9.5.3); the real Jordan block $\mathbf{C}_k(\lambda)$ (Definition 9.4.1); a circulant (Definition 11.9.1). Plain C and C_b^a are contractions (Definition 14.4.1) and C_{ij} a cofactor (Definition 6.6.1); blackboard $\mathbb{C}$ is the complex numbers	the companion matrix always carries its polynomial, the Jordan blocks their size and eigenvalue; a bare $\mathbf{C}$ in Chapter 22 is a circulant
$\mathbf{W}, W$	the Wielandt matrix $\mathbf{W}_n$ (Theorem 18.5.6); the unnormalized transform matrix $\mathbf{W}_n$ of Chapter 23 §12; the accumulated orthogonal factor $\mathbf{W}_k$ of Chapter 23 §08. Plain W is a subspace, $W(\mathbf{A})$ the numerical range (Definition 19.2.1), and W_n the zero-sum matrices of Chapter 18 §07 (Lemma 18.7.1)	the numerical range always carries its matrix as an argument; the subspace never carries one
Θ, θ	the isomorphism $\mathbb{C} \rightarrow \mathbb{R}^2$ (Example (The Standard Isomorphisms)); the basis-dependent $V \rightarrow V^*$ (Theorem 4.3.2); the natural isomorphism $V^* \otimes W \rightarrow \mathcal{L}(V, W)$ (Theorem 14.3.9); the diagonal matrix of principal angles (Definition 19.8.1). Lowercase: the angle between vectors (Definition 10.1.5), the choice-of-origin map θ_O (Proposition 21.1.3), the Ritz values $\theta_1, \dots, \theta_k$ (Definition 23.6.2)	the principal-angle Θ appears only inside $\sin \Theta$ and $\cos \Theta$; lowercase θ is an angle unless the section says otherwise, and the choice-of-origin map carries the point O

Symbol or word	The meanings	How to tell
Π, π	bold Π is the limit of a convergent stochastic matrix (Theorem 9.10.4); plain Π is a plane in Chapter 21 §§04, 08, 09 and the quotient map $\text{GL}(V) \rightarrow \text{PGL}(V)$ in §07; π is the quotient map $X \rightarrow X/\sim$ of a set (Definition 0.5.5), a quotient homomorphism of algebras (Definition 22.2.2) and, locally, a slot projection	the typeface, and that no section of Chapter 21 uses both meanings
Φ, Ψ	a symmetric gauge function and its dual (Definition 20.3.1, Definition 20.3.3); the two isomorphisms of Chapter 22 §09 (Definition 22.9.5)	the argument: a gauge eats a real vector, Chapter 22's maps eat an element of a group algebra
Ω, ω	plain Ω_n is the doubly stochastic matrices (Definition 17.8.4), bold Ω_{2m} the standard alternating matrix (Definition 13.10.2). Lowercase ω is a root of unity (Definition 18.5.2, Definition 23.12.1), an angular frequency (Definition 23.9.2), the exponent of matrix multiplication (Definition 23.13.2), and a differential form being wedged (Definition 14.10.1)	the typeface for Ω ; for ω , the section, each of which declares its use
ρ	spectral radius (Definition 15.4.1); a representation of a group or an algebra (Definition 22.7.1)	the argument: a matrix for the radius, a group element for the representation
σ	a permutation (Definition 6.3.2); a singular value $\sigma_i(\mathbf{A})$ (Definition 12.8.1)	the singular value always carries an index and a matrix
λ, μ	a scalar or an eigenvalue throughout; also the linear-part homomorphism $E(n) \rightarrow O(n)$ (Definition 21.4.1) and the Fiedler value μ_2 (Definition 23.11.2)	Chapter 21 §04 is the one place λ is not a scalar, and it says so
$ \cdot $	modulus of a complex number (Definition 0.6.2); cardinality of a set; the matrix absolute value $(\mathbf{A}^* \mathbf{A})^{1/2}$ (Definition 12.2.1); the entrywise absolute value in Chapter 18 (Definition 18.1.1)	what sits inside. Chapter 18 states its override, and the Chapter 12 meaning does not appear there
Λ	plain $\Lambda^k V$ is an exterior power (Definition 14.8.2); bold $\Lambda(\lambda)$ is a rotation-scaling block (Definition 9.4.1)	the typeface, and the block always carries its eigenvalue
δ	Kronecker delta δ_{ij} (Definition 0.9.3); a point mass δ_g in a group algebra (Definition 22.1.5); a basis element δ_s of a free vector space (Definition 14.2.3); one rounding error (Definition 23.1.1)	the Kronecker delta always carries two numerical subscripts
$\perp$	orthogonal complement $U^\perp$ (Definition 10.3.1); the complement $U^{\perp\beta}$ for a bilinear form (Definition 13.1.3); orthogonal direct sum $U \perp W$ of a form (Definition 13.7.3); a polarity $S^\perp$ in projective space (Theorem 21.8.3)	the unadorned $U^\perp$ is always the inner-product one

Symbol or word	The meanings	How to tell
$\cong, \sim, \simeq, \approx$	isomorphic spaces (Definition 3.5.3) or, between spaces carrying forms, isometric (Definition 13.8.1); similar matrices (Definition 3.7.2); congruent matrices (Definition 13.2.1); equivalent over $F[x]$ (Definition 9.7.2), or “approximately equal” in prose	what stands on either side; the four relation symbols are never interchanged
brackets	$[\mathbf{v}]_{\mathcal{B}}$ a coordinate vector (Definition 1.5.2); $[T]_{\mathcal{B}}^{\mathcal{C}}$ a matrix of a map (Definition 3.6.1); $[\mathbf{A}, \mathbf{B}]$ a commutator (Definition 7.5.1); $[\mathbf{x}, \mathbf{y}]$ a segment (Definition 17.1.1); $[\mathbf{x}]$ a projective point (Definition 21.6.2) and $[T]$ a projectivity (Definition 21.7.1); $F[x]$ and $F[G]$ rings and algebras (Definition 0.8.2, Definition 22.1.5)	the number of arguments and whether they are bold: a segment takes two bold vectors, a commutator two bold matrices
the superscript of $[\cdot]_{\mathcal{B}}^{\mathcal{C}}$	for a linear map it is the output basis, $[T]_{\mathcal{B}}^{\mathcal{C}}$ (Definition 3.6.1); for a bilinear form it is the basis read into the second slot , $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ (Definition 13.1.2)	what sits inside the brackets. The two readings are the same rule — the basis on the second slot — since a map’s second slot is its output; a form on $V \times V$ always gives both slots the same basis, so its two labels agree, and neither is ever left off
T	a linear map throughout; the transpose $\mathbf{A}^{\top}$; Chebyshev polynomials T_n (Definition 23.7.4); the tensor algebra $T(V)$ (Definition 14.6.2) and mixed tensors $T_q^p(V)$ (Definition 14.4.2); a T -transform (Definition 20.1.1)	the tensor algebra is upright and takes a space; the Chebyshev polynomial and the T -transform carry subscripts
P, p	orthogonal projection P_U (Definition 10.3.2); Legendre polynomials P_n (Definition 10.10.1); a point of an affine space (Definition 21.1.1); the nearest-point map P_C (Theorem 17.3.1); a face P_J (Proposition 21.3.1); the even-odd permutation $\mathbf{P}_n$ (Definition 23.12.2); characteristic polynomial p_T (Definition 6.10.1); a gauge p_K (Definition 17.4.1)	a subspace subscript means a projection, an integer subscript a polynomial; bold means a matrix
Z	the cyclic subspace $Z(\mathbf{v}; T)$ (Definition 9.5.1); the center $Z(A)$ of an algebra (Definition 22.1.6); the zero set $Z(q)$ of a quadratic form (Definition 21.10.1); a Z-matrix (Definition 18.9.1)	the argument decides: a vector and an operator, an algebra, or a form
$\mathbf{E}, E$	the perturbation in $\mathbf{A} + \mathbf{E}$; a matrix unit $\mathbf{E}_{ij}$ (Example (Standard Bases)); an elementary matrix (Definition 2.4.1); an eigenspace $E_{\lambda}(T)$ (Definition 8.2.2); the Euclidean group $E(n)$ (Definition 21.4.1)	subscripts again: two indices give a matrix unit, an eigenvalue gives an eigenspace, a bare $\mathbf{E}$ is the perturbation

Symbol or word	The meanings	How to tell
D , <i>D</i>	a diagonal matrix (Definition 0.9.4); the elementary scaling $\mathbf{D}_i(c)$ (Definition 2.4.1); the twiddle matrix $\mathbf{D}_m$ (Definition 23.12.2); the degree matrix in $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (Definition 23.11.1); a Gershgorin disc $D_i(\mathbf{A})$ (Definition 19.1.1); density matrices $\mathcal{D}_n$ (Definition 17.8.3)	the Gershgorin disc is plain italic, since the bold letter is a diagonal matrix
M , <i>M</i>	a block matrix, as in the Schur complement $\mathbf{M}/\mathbf{A}$ (Definition 7.3.1); the mass matrix (Definition 23.9.2); the plane reflection $\mathbf{M}_\theta$ (Theorem 10.7.5); a minor M_{ij} (Definition 6.6.1); the space $M_n(F)$ (Definition 0.9.1); an M-matrix (Definition 18.9.2)	plain italic with two indices is a minor; $M_n(F)$ always carries its size and field
<i>u</i>	the unit roundoff (Definition 23.1.1); bold u is a vector	the typeface; Chapter 23 §01 warns about the pair
<i>h</i>	the period of an irreducible non-negative matrix (Definition 18.5.2); a support function h_K (Definition 17.4.2); a mesh width in Chapter 18 §09	the support function always carries its convex set
simple	a simple algebra (Definition 22.3.1); a simple A -space (Definition 22.3.3); a simple eigenvalue, of multiplicity one (Definition 19.11.1); a simple tensor (Definition 14.2.2)	what the word is applied to
equivalent	equivalent matrices $\mathbf{B} = \mathbf{QAP}$ (Definition 3.8.1); equivalence over $F[x]$ (Definition 9.7.2); equivalent norms (Definition 15.2.1); equivalent representations (Definition 22.7.3); affine equivalence of quadrics (Definition 21.5.2)	the objects being compared
positive	$\mathbf{A} > 0$ entrywise, every entry positive (Definition 18.1.1); $\mathbf{A} \succ 0$ positive definite (Definition 12.1.1)	the symbol: $>$ is entrywise, $\succ$ is the Loewner order. A positive matrix need not be positive definite, nor conversely
radical, contraction	radical of a bilinear form (Definition 13.1.3) or of an algebra (Definition 22.3.5); contraction of tensor slots (Definition 14.4.1) or a matrix of norm at most one (Definition 20.8.1)	the argument, in both cases
signature	the pair (n_+, n_-) , and also the single integer $n_+ - n_-$	Definition 13.5.1 says which is meant where the word appears